

Weekly Report: [W13]

General Information

- **Group ID:** 52
- **Group Name:** Group 52
- **Class:** 25A01
- **Project Name:** Super Mario Game
- **Date range:** 2026-08-30 – 2026-09-05 (Sunday–Saturday; partial week — this report is compiled and reported as of 2026-09-02, three days before the week closes, per the submission-sweep plan's explicit instruction. Commits after 2026-09-02 18:54, this report's own compile time, are not reflected and belong in a later update.)
- **Github Repository:** <https://github.com/ndmhuy/SuperMarioGame.git>

Tasks Completed This Week

This is a partial week, reported mid-flight. Today is 2026-09-02; W13 runs through 2026-09-05. Per the pre-flight checklist (docs/REPORT_RULES.md §2.1): `git fetch --all` was run first; `dev` is even with `origin/dev` (`git rev-list --left-right --count dev...origin/dev` → `0 0`), so nothing below is stale relative to the remote.

Commit count: `git log --since='2026-08-30' --until='2026-09-02 23:59:59'` on `dev` returns **175 commits** — matching the submission-sweep plan's count exactly. `--all` returns **179**: the extra 4 sit on `A/fix/duplicate-playingstate-exit`, an unmerged branch (see "Unmerged work" below). Per day on `dev`: **142** on 2026-08-31, **14** on 2026-09-01, **19** on 2026-09-02 (through the time of writing).

Per-member split (`git log --format='%an' dev` over the window): **172** commits by Nguyễn Đình Minh Huy, **3** by Trần Gia Huy (git identity `FubuGold`). Member B's share is small and is reported as such, not padded.

Branches merged into `dev` this week (30 merge commits; every one of the following was branched off `dev`, implemented, and merged back the same session — none touched `main` except the one user-directed merge noted below): `A/docs/report-architecture-figure-and-full-uml`, `A/docs/detailed-uml-full-attributes-methods`, `A/docs/spec-feature-audit`, `A/fix/audit-31-08-defect-batch` (R1), `A/fix/animation-manager-`

disposition (R2), A/fix/no-audio-device-degrade (R3), A/fix/scoped-subscription-adoption (R4), A/refactor/collision-dispatch (R5), A/docs/hygiene-and-descope-addendum (R6), A/feature/wire-dormant-vfx-sfx (R7), A/feature/load-game-menu (R8), A/feature/campaign-population-pass (R9), A/fix/tests-honour-assertions + A/build/object-library + A/fix/hermetic-tests (R11), A/feature/attract-mode (R10), A/fix/level-completion-and-camera-defects, A/docs/submission-package-update, A/feature/endless-mode-and-solvability, A/fix/boss-encounter-batch (R15), A/fix/sublevel-tile-batch (R16), A/fix/player-enemy-visual-batch (R17), A/fix/endgame-audio-batch (R19), A/docs/audit-status-reconciliation, A/verify/full-playtest-pass (R12), A/fix/remaining-defects-d29-d30-exit (R20), A/fix/boss-falls-out-of-world, A/release/defect-batch-r21 (R21 wave 1), and A/fix/versus-camera-and-axes (R21 wave 2, merged as "Lane M").

main was fast-forward-merged only twice at the user's explicit direction (2026-08-31 09:45 and 10:00) and is **not** touched again after that — per AGENTS.md directive 1, promoting main further stays the user's decision.

25125083 - Nguyễn Đình Minh Huy (Member A - Engine, Physics & Player/Item Entities) — 172 commits

Branch: A/docs/submission-package-update, A/docs/report-architecture-figure-and-full-uml, A/docs/detailed-uml-full-attributes-methods → merged to dev (2026-08-31 morning)

- **Task 1: Submission package v1:** Rebuilt the full submission set from SPEC.md, git history and logs/agent_history.log rather than from memory: features_list.md grew from 40 generic items to 97, each individually checked against running source and marked reachable-from-gameplay or editor-only; testing_tasks.md rebuilt as a per-feature manual test checklist; Member_Contributions.md / .xlsx reconstructed as a 118-task, 214.5-hour breakdown from all 7 prior weekly reports and the git log, reporting the raw 82/18 commit split alongside the closer 62/38 task/hour split rather than only the flattering number; AI_Usage_Declaration.md corrected to name Claude Code as the tool actually used this session. Verified by rebuilding the report toolchain and reading the rendered PDF pages back (pdftotext) after catching that a first build had used stale content.
- **Task 2: Report architecture figure and full-attribute/method UML:** Replaced the ASCII-art architecture diagram in reports/report_content.py with a real figure per the user's explicit correction ("it should be a real figure not those hard to read code"), then extended SuperMarioGame/tools/gen_class_diagram.py so the appendix UML diagrams show every attribute, method and visibility marker instead of a bare class-name box — the change the user asked for specifically so a reader could verify encapsulation/abstraction/

polymorphism by eye. Rebuilt `class_diagram.md` and all three report editions (HTML, LaTeX PDF at `Report/SuperMarioGame/Group52_SuperMarioGame_Report.pdf`, and the `submission_documents/` copies).

Branch: `A/fix/level-completion-and-camera-defects` → merged to `dev` (2026-08-31 08:45)

- **Task 3: Three level-completion/camera defects:** See Issue 1 below. Widened `level_2.json` / `level_3.json` from 200 to 208 tiles to restore the castle end-buffer both already had elsewhere; shrank Boom Boom's `arenaW` from 11 to 7 so it no longer overlaps the flagpole; added a boss-alive gate to `PlayingState`'s `LevelComplete` subscriber; reset the camera's scroll mode and position in `PlayingState::setupTestScene()` on every level load; reordered `render()` so the player always draws after the general entity loop instead of relying on insertion order. ctest 13/13, plus a live run of `tests/scripts/verify_1_3_camera_and_visibility.txt`.

Branch: `A/feature/endless-mode-and-solvability` → merged to `dev` (2026-08-31 09:10)

- **Task 4: Endless Mode + `LevelSolvability` oracle:** Implemented Endless Mode end to end — `PlayingState` gained an `isEndless` flag and `extendEndlessLevelIfNeeded()`, called once per frame, that generates each chunk into an isolated tile map/entity list and splices it in with a flat safety bridge across the seam; a menu row ("PLAY ENDLESS" on the Procedural page in `MenuState.cpp`) constructs a real `PlayingState`, not a harness. Wrote `Utils/LevelSolvability (.cpp)` from scratch: a column-reachability BFS bounded by the game's own jump/run constants, with `MovingPlatform` / `FallingPlatform` counted as standable ground at their own column. `MapGenerator::generateSolvable()` wraps `generate()` and retries with a new seed on failure; every procedural path (procedural mode, Endless chunks) now calls it instead of raw `generate()`. Investigated the two abandoned side branches (`A/rl-neural-policy`, `A/mapgen-gan-plan`) first via a read-only subagent and did **not** port either — the neural policy never worked (its own log documents it converging on "stand still"), and the GAN pipeline is inseparable from a vendored C++20 tensor library — but adopted the *idea* of a solvability gate as this from-scratch reimplement. Verified live: `tests/scripts/verify_endless_mode_long.txt` showed the distance counter climbing, a real "appended chunk 1, tilemap now 300 tiles wide" log line, and a seamless crossing of the chunk boundary.

Branch: `A/docs/spec-feature-audit` → merged to `dev` as `021c1b3` (2026-08-31 11:15)

- **Task 5: Full SPEC/feature audit:** Wrote `docs/issues/spec_feature_audit_2026-08-31.md`, auditing every SPEC.md claim against the

running code and folding in 13 playtest-reported defects (D16–D28) into an R1–R19 remediation plan, each item scoped, given an acceptance bar, and a model/effort tier. This document is the ground truth every R-numbered task below traces back to.

Branches `A/fix/audit-31-08-defect-batch (R1)`, `A/fix/animation-manager-disposition (R2)`, `A/fix/no-audio-device-degrade (R3)`, `A/fix/scoped-subscription-adoption (R4)` → merged to `dev (2026-08-31 11:48–12:24)`

- **Task 6 (R1):** Fixed `SoundManager::playLevelBGM` 's index-2 case resolving to "underwater" instead of "castle" (`SoundManager.cpp`); made `Enemy` / `Spiny` despawn against the current level's own `TileMap` height instead of the fixed 720px screen constant; surfaced `MapGenerator::generateSolvable` 's previously-discarded bool return into `PlayingState::m_lastLevelUnverified` with a dev-panel warning; registered the orphaned `verify_sound_visual.cpp` harness in `CMakeLists.txt` .
- **Task 7 (R2):** Deleted the dead `AnimationManager` singleton (`Graphics/AnimationManager.hpp` / `.cpp` , both removed) after confirming zero reachable references remained; the real animation path (`Character::setupCharacterAnimations` , per- `Enemy` `setupAnimations()`) never used it.
- **Task 8 (R3):** See Issue 3 below — added `SoundManager::m_audioAvailable` , probed once via `sf::PlaybackDevice::getDefaultDevice()` , so a device-less machine degrades to silent instead of `SIGABRT` .
- **Task 9 (R4):** Migrated every raw `EventBus::SubscriptionId` outside `EventBus` itself (14 in `PlayingState` , plus `AchievementManager` , `StatisticsTracker` , `Camera`) to `EventBus::ScopedSubscription` ; see Issue 4 below for the static-destruction-order `SIGSEGV` this surfaced and fixed.

Branch: `A/docs/hygiene-and-descope-addendum (R6)` → merged to `dev (2026-08-31 12:34)`

- **Task 10:** `git rm --cached` on `.member_profile.json` ; generated the missing `docs/Group52_08/52.pdf` ; added a "Descope Addendum" (SPEC.md §21) listing 16 explicitly descope items, correcting a fabricated "measured benchmark" claim in the process; documented the `Entity.hpp` / `Character.hpp` friend-sprawl (12 friends each) as a deliberate, unmeasured design trade-off rather than restating a false performance claim.

Branch: `A/refactor/collision-dispatch (R5)` → merged to `dev (2026-08-31 18:50)`

- **Task 11:** See Issue 5 below. Reduced `CollisionResolver.cpp` 's `dynamic_cast` count from 28 to 0 via virtual contact hooks on `Entity` and an ordered `EntityCategory` -pair switch, honouring the Open/Closed constraint (no new type-tag enum).

Branch: `A/feature/campaign-population-pass` (R9) → merged to `dev` (2026-08-31 19:02)

- **Task 12:** Placed the six implemented-but-unused enemy types (`KoopaParatroopa` , `Boo` , `BulletBill` , `Thwomp` , `ChainChomp` , `Lakitu`) across the seven shipped levels, added a hidden block per main level, rebalanced per-level enemy counts against the Easy/Normal/Hard modifiers, and extended `verify_regressions.cpp` with a permanent `LevelSolvability` case run over all seven levels.

Branch: `A/feature/wire-dormant-vfx-sfx` (R7) → merged to `dev` (2026-08-31 20:33)

- **Task 13:** Wired star-kill/player-death floating overlays and their particles, surface footsteps, combo SFX escalation, a per-row menu navigation cue, the versus-mode camera clamp, and console Tab-completion — seven previously-built-but-dormant systems (F2 + F4 + F6 from the audit), each re-verified against `dev` before wiring since the audit's own dormant-feature inventory had two entries backwards.

Branches `A/fix/tests-honour-assertions` , `A/build/object-library` , `A/fix/hermetic-tests` (R11) → merged to `dev` (2026-08-31 19:27–20:44)

- **Task 14:** See Issue 6 below (assertions compiled out under `NDEBUG`). Also introduced two CMake `OBJECT` libraries (`AppObjectsRelease` , `AppObjectsTest`) so 23 `verify_*` targets compile the ~133 `APP_SOURCES` once instead of once each (previously ~2,900 `.o` files across the build tree); and added a permanent CTest guard that fails if any test touches the real `SuperMarioGame/saves/` directory, verified by deliberately breaking a test and watching the guard fail before reverting.

Branch: `A/feature/load-game-menu` (R8) → merged to `dev` (2026-08-31 21:27)

- **Task 15:** Added a LOAD GAME row to `MenuState` opening a 3-slot picker that renders `Serializer::getSlotPreview` (character/level/score/star coins/play time), reusing the exact `PlayingState::loadFromSlot` path the dev panel already used rather than forking a second one.

Branch: `A/feature/attract-mode` (R10) → merged to `dev` (2026-08-31 22:10)

- **Task 16:** Implemented main-menu attract mode (SPEC 10.2): after 30s idle on the menu, plays a bundled ~30s World 1-1 demo recording via the existing `ReplayRecorder` playback path, dismissed instantly by any key, guarded against ever running during an active game or in Endless/versus.

Branch: `A/verify/full-playtest-pass` (R12) → merged to `dev` as `7bc1dd9`
(2026-08-31 23:05)

- **Task 17:** Full scripted playtest pass — a complete campaign walkthrough through both bosses, P-Switch/POW/axe/100-coin 1-UP, a 2P versus death, Easy vs. Hard on the same level, a live key rebind driving play, and a 60s FPS trace. Reconciled the D14 Windows-crash log discrepancy and filed two new defects (D29, D30) rather than fixing them inline, per the audit's own "verify before improvising" discipline.

Branch: `A/fix/remaining-defects-d29-d30-exit` (R20) → merged to `dev` as `3237ad6` (2026-09-01 09:16)

- **Task 18 (D29):** A death+respawn near an active boss silently deleted it — fixed in `PlayingState.cpp` (commit `ec33965`).
- **Task 19 (D30):** LOAD GAME resumed the wrong level relative to what a save slot actually recorded — fixed (commit `bab8b31`).
- **Task 20:** Re-added `PlayingState::exit()`'s double-call guard (commit `abc0870`) after the resubscribe regression test (`tests/scripts/verify_r4_resubscribe.txt`) was found to only print the log line without actually observing teardown; strengthened in commit `6e4868d` so the test genuinely watches `exit()` run once.

Branch: `A/fix/boss-falls-out-of-world` → merged to `dev` (2026-09-01 11:05, commit `95521a8`)

- **Task 21:** See Issue 7 below. Added `Boss::returnToArenaSpawn()` so a boss pushed out of its arena's bounds is teleported back instead of falling forever off-screen.

Branch: `dev` direct (2026-09-01, 15:31–17:55 session)

- **Task 22:** Ten gameplay/visual fixes reported live by a teammate — see the Member B section below; this session's own commits landed as `384250f`, authored under Member B's git identity, and are attributed there rather than duplicated here.

Branch: `A/release/defect-batch-r21` (R21, wave 1 — 15 items) → merged to `dev` as `97fc43a` (2026-09-02 14:05, tip `4474ac6`)

- **Task 23: One asset tree.** See Issue 8 below — `tools/generate_game_levels.cpp` stopped double-writing every level, a stray root-level `assets/` tree was deleted after being backed up outside the repo, and `tests/guard_asset_single_source.cpp` now fails the build if a second tree reappears.
- **Task 24: Real map editor.** New `Core/EditorState` (.cpp) replaces the old palette that hand-rolled 16 type names and silently placed nothing for 24 of 40 buttons — see Issue 9 below. The open file is named in the chrome as BUILT-IN/CUSTOM/NEW, custom levels

save under `assets/levels/custom/` with the absolute path shown, a CUSTOM LEVELS menu page plays them, and F5 round-trips into a playtest and back.

- **Task 25: One entity registry.** `Entities/EntityCatalogue (.cpp)` is one table row per entity type; `tests/verify_r21_entity_registry.cpp` walks `EntityType::0..Count` so a forgotten row fails the build instead of becoming a dead palette button.
- **Task 26: Dynamic lighting, for real.** Built `Graphics/LightingRenderer (.cpp)` with a real GLSL 1.20 fragment shader (`assets/shaders/radial_light.frag`): theme-driven darkness, fireballs rendered as moving lights, a 100s raised-cosine day/night cycle, and a graceful no-GPU fallback (verified by moving the `.frag` out of the tree and re-running). This resolves a standing contradiction where `SPEC.md` 's 110/115 rubric credited "GLSL shader effects" that did not exist.
- **Task 27: Recording cheats.** `Core/DebugCheats` — immortality now *rescues* the player onto solid ground in the column they fell from (mirroring `Boss::returnToArenaSpawn()`) instead of merely suppressing the void-kill, which would otherwise drop the player into an endless fall. A cheated run is permanently tainted: no high score, no achievement credit.
- **Task 28: Procgen far-chunk entities.** Gave `Entity` a virtual `translate()` so `MovingPlatform / PiranhaPlant / FallingPlatform / Boss` stop caching spawn positions in chunk-local coordinates and teleporting back on frame 1 in Endless Mode; added a real boss arena every 5 chunks and a 45% dead-prefab margin fix. Also closed the `LevelSolvability` vacuous-truth defect for the castle/underground themes (top-down `groundRowAt` scan hit a solid ceiling at row 0 and the BFS goal excluded the exit apron) in commit `081c4d6` .

Branch: `dev` *direct* — *R21 wave 2 (10 items)* — *commits* `1649618` , `8a7cd15` ,
merged as `e97b7da` (2026-09-02 17:40)

- **Task 29:** See Issue 10 below (Bowser's fireball stagger). Rebuilt pipe rendering to share one `PipeRenderer::cellArt` between the entity and tilemap paths, giving `Pipe::EntryMode` real meaning (down through a rimmed mouth vs. up out of an L-bend); made 2–3 bridge axes (by difficulty) each individually required to drop the bridge; gave Bowser a `beginLavaDeath()` fall-and-drain sequence instead of an instant vanish; made Lakitu drop a Fire Flower so the fight has a guaranteed route to the fireballs its own stagger requires; gave the 2P survivor the camera and an "OUT" badge for the eliminated player (see Issue 11 below); added slot choice to SAVE GAME and delete-save to LOAD GAME (`Serializer::deleteSlot` already existed with no player-reachable caller). ctest 28/28 on the merged `dev` .

Branch: `dev` *direct* — *commit* `6af4f8e` (2026-09-02 17:18, *tip of this week's* `dev`)

- **Task 30:** See Issue 12 below (flagpole touched during a boss fight). ctest 28/28 → 29/29.
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25125084 - Trần Gia Huy (Member B - Input, Sound & Gameplay Systems) — 3 commits

Branch: `dev` *direct* — **commit** `384250f` *"feat: fix sublevel warps, pipe rendering, crouch controls, AI movement, and boss arena enclosure"* (2026-09-01 17:55)

- **Task 1: Ten teammate-reported gameplay/visual fixes:** Replaced the undersized `castle_end` sprite (112×176) with the full `castle_0` (148×176) in `Entities/Castle` (.cpp), correcting `WIDTH_TILES` to `4.625f`; extended `bonus_1.json`'s plateau at rows 16–20 so the flagpole and castle seat correctly; added `hasStarPower()` immunity to `Player::takeDamage()` and wired an authentic 60fps Star-flicker palette into `SpriteColorFilter::getMarioStarPaletteColor`; corrected `Entity::drawSprite`'s flipped-Y origin so it pivots on the horizontal centre line instead of drifting; enabled secondary jump keys in `InputManager::isActionHeld()` so Peach can float with either Space or W (P2: Up or RShift).
- **Task 2: Sub-level pipe teleports and crouch/AI rework**, folded into the same commit: sub-level exit pipes now land the player beside the pipe on solid ground instead of teleporting into a wall; decoupled `S` /Down from Ground Pound so `S` exclusively crouches (Ground Pound moved to `Q` /Slash); halved Thwomp's slam speed and narrowed its detection column to the exact column beneath it; rewrote `TetheredChaseStrategy` so Chain Chomp actively tracks, faces and strains toward the player with an elastic-leash rebound instead of sitting inert; added 1-tile enclosure walls at the World 1-3 bridge boundaries (x=180, x=189) and synced the camera to the boss arena lock so it snaps correctly on Bowser's defeat.

Branch: `dev` *direct* — **commit** `60becee` *"chore: video requirements"* (2026-09-01 23:24)

- **Task 3:** Authored `submission_documents/demo_video_requirements.md`, an 8-section checklist mapping every planned demo video to what it must show.

Branch: `dev` *direct* — **commit** `1a7a6db` *"feat: fix minimap M toggle, single death animation, metal sfx volume, and Thwomp mechanics; export demo video checklist"* (2026-09-01 23:32)

- **Task 4:** Fixed the minimap's `M` key toggle in `InputManager.cpp`, a duplicated player-death animation firing twice in `Player.cpp`, an over-loud metal-footstep SFX volume, and further tuned Thwomp's slam/rise timing (elevated in World 1-3, faster recovery rise).

Unmerged work — not counted as delivered

`A/fix/duplicate-playingstate-exit` carries 4 commits (`f169a6c` , `d6375dd` , `cdcb4f9` , `1b3a2c9` , 2026-08-31/09-01) that re-add a `PlayingState::exit()` double-call guard against an **old** checkout of `PlayingState` predating `dev`'s attract mode, `ScopedSubscription` tokens, and Load Game slot work. The branch's own last commit message says it plainly: *"do not merge into dev as-is, it would revert unrelated features. Superseded by `A/fix/duplicate-playingstate-exit` (`f169a6c`) and `dev`'s own current implementation."* `dev` already carries an equivalent, more complete guard via R20 (Task 20 above). This branch is not part of the delivered week's work and is recorded here per `docs/REPORT_RULES.md`'s "unmerged — on `<branch>`" convention rather than silently ignored.

AI Usage Declaration

- Used AI (Claude Code) to root-cause every defect in this report's Issues & Resolutions section *before* writing a fix — a dedicated read-only Explore/diagnosis pass preceded code changes for the level-completion/camera bug, the R21 asset-tree dual-write, the map-editor palette failure, and Bowser's fireball-stagger bug, per AGENTS.md directive 2 (root-cause before code).
 - Used AI to design the `LevelSolvability` column-reachability BFS and its integration into `MapGenerator::generateSolvable()` , and to evaluate (and reject) porting the two abandoned AI side-project branches (`A/rl-neural-policy` , `A/mapgen-gan-plan`) into Endless Mode.
 - Used AI to implement the GLSL 1.20 fragment shader (`radial_light.frag`) and its C++ integration in `LightingRenderer` , including the no-GPU fallback path.
 - Used AI to diagnose the `EventBus::ScopedSubscription` static-destruction-order `SIGSEGV` (Issue 4) and the `NDEBUG`-strips-`assert()` vacuous-test defect (Issue 6), both requiring build/runtime-order reasoning beyond a simple grep.
 - Used AI to draft and regenerate the submission package (`features_list.md` , `Member_Contributions.md/.xlsx` , `AI_Usage_Declaration.md`), the LaTeX/HTML report editions, and the full-attribute/method UML diagrams, cross-checked against source rather than written from memory.
 - Antigravity was the primary AI tool for the bulk of this project's earlier phases per the corrected `AI_Usage_Declaration.md` (2026-08-31 10:00 session); Claude Code is named as the tool actually used for this week's sessions specifically.
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Tasks Planned for Next Week

- **Nguyễn Đình Minh Huy:** Complete the remaining W13 days (2026-09-03 ... 09-05) per the submission-sweep plan (`docs/issues/submission_sweep_plan_2026-09-02.md`): Phase 3 evidence capture (screenshot catalogue, demo-video shot lists), Phase 4B/4C document refresh (exhaustive features list, member contributions generator), and the start of Phase 5's report rewrite (design-pattern reasoning, SOLID walkthrough, LaTeX layout fix for the overlapping-text defect). Not fixable this week and stated as such per the plan: the Windows-platform claim (no Windows CI), the third HARD bridge axe standing behind the arena enclosure post at (189,19), and `MovingPlatform::update()` never calling `Block::update()` .
- **Trần Gia Huy:** Continue teammate playtesting and file further live-session defects the same way Task 22/1 above were reported (direct observation, not code review); record the planned demo videos against the checklist in `submission_documents/demo_video_requirements.md` .

Issues & Resolutions

Issue 1: Level-Completion Gate Absent, Stale Camera Mode, and Player Z-Order Inversion

- **Problem:** Three defects shared one investigation. (a) `level_2.json` / `level_3.json` placed their castle with only 0.5 tiles of buffer to the level's right edge (`level_1.json` / `bonus_1.json` have 8.5), and the level-complete sequence gave the player no walk-to-the-door beat on any stage, so the end-of-level walk drove the player into the castle wall. (b) Boom Boom's arena (`arenaX 185, arenaW 11`) spanned past the flagpole at `x=193` , so the "no escape until defeated" clamp still let a player reach a touchable flag with the boss alive. (c) `PlayingState::setupTestScene()` never reset the camera's `ScrollMode` or position on a new level load, so a `Locked` mode left over from a boss arena carried into the next level, and `adoptPlayer()` inserting the player at the front of `m_entities` meant `render()` 's draw-in-list-order painted later entities (bosses) over the player, making them invisible.
- **Resolution:** Widened `level_2.json` / `level_3.json` from 200 to 208 tiles (only the spans already touching the old edge) to restore the castle buffer; shrank Boom Boom's `arenaW` from 11 to 7, leaving a 1-tile gap before the flagpole (mirroring `level_3.json` 's own Bowser arena); added a code-level gate in `PlayingState` 's `LevelComplete` subscriber that refuses completion while `m_activeBoss` is active, independent of level data; gave the level-complete hold a real "walk to the castle door" sequence driven by

`moveLeft()` / `moveRight()` ; made `setupTestScene()` unconditionally reset the camera to `Free` mode and snap it to the new level's spawn point on both the success and fallback paths; reordered `render()` to draw `m_player` / `m_player2` after the general entity loop. Verified live via `tests/scripts/verify_1_3_camera_and_visibility.txt` (confirmed 1-3 opens on the player's own spawn corner and the player stays visible over foreground entities) plus 13 new/strengthened ctest cases; the scripted run did not survive far enough to reach Bowser himself, so that specific frame rests on the ctest case and code review only — noted honestly in `docs/verification/README.md` rather than claimed as observed.

Issue 2: One Bug Reported as Three — the Dual-Write Asset Tree

- **Problem:** `tools/generate_game_levels.cpp` wrote every level twice — once relative to the working directory and once to `"../" + path` — creating an untracked second `assets/levels/` tree at the repository root. `ResourceManager::resolvePath` tries the bare relative path **first**, so which tree the game read depended entirely on the launch directory: from `SuperMarioGame/` the tracked tree (correct); from the repo root, the stray tree (stale). The stray tree was frozen at 2026-08-31 12:35, **nine minutes before** the two commits (`309cla5` , `cc6a32d`) that fixed the exact defects it still contained (P-Switch deleting sub-level floors; a 1-column "half" entrance pipe), and had flattened every level's `theme` to `overworld` . Three separately-reported bugs — half pipes, a broken 1-1 Sub P-Switch, and "two sources of truth for asset detail" — were one bug: whoever filed a report and whoever tried to reproduce it were not necessarily running the same files.
- **Resolution:** Deleted the dual write from `generate_game_levels.cpp` ; gitignored `/assets/` at the repository root (anchored, not recursive, so the tracked `SuperMarioGame/assets/` is untouched); added `tests/guard_asset_single_source.cpp` , which fails the build if a second tree reappears and also asserts the canonical tree is intact (so deleting the *right* tree cannot silently pass). The stray tree was backed up outside the repository before removal and the deletion was explicitly approved by the user, per AGENTS.md directive 7 (never discard uncommitted work without confirmation).

Issue 3: Game `SIGABRT` s on Startup With No Audio Playback Device

- **Problem:** `SoundManager::getInstance()` built a pool of 64 real `sf::Sound` objects at construction time regardless of hardware. On a machine (or CI runner) with no sound card, SFML's `miniaudio` backend fails to open a playback device and drives the process to `SIGABRT` before the first test assertion runs — the reason `.github/workflows/ci.yml` had been installing a dummy ALSA device as a workaround instead of the game handling the absence itself.
- **Resolution:** Added `SoundManager::m_audioAvailable` , probed exactly once in the constructor via `sf::PlaybackDevice::getDefaultDevice().has_value()` (confirmed

safe against SFML 3.0.2's vendored source — this call only enumerates and logs, never throws). When unavailable, the `sf::Sound` pool is skipped entirely and `playSound()` / `playMusic()` become no-ops; `stopMusic()` / `pauseMusic()` / `resumeMusic()` were deliberately left unguarded since SFML already no-ops those on a never-opened `sf::Music`. Verified with two manual runs — unmodified (reaches `MenuState` silently) and with the probe hardcoded false (reaches `MenuState` with exactly one "running silent" warning, no abort) — since macOS provides no way to actually remove the audio device for a true device-less test.

Issue 4: `SIGSEGV` at Program Exit After Migrating to `ScopedSubscription`

- **Problem:** Converting `AchievementManager` / `StatisticsTracker`'s subscriptions from raw `EventBus::SubscriptionId` to `std::vector<EventBus::ScopedSubscription>` turned `verify_save_load`'s `testTrackerAndAchievements` into a reliable `SIGSEGV` (ctest exit 139) at program exit. `EventBus::ScopedSubscription`'s destructor calls `EventBus::getInstance().unsubscribe()`, and a vector of them owned by a Meyers-singleton destructs at static-destruction time; function-local statics destruct in reverse order of first construction, and which of `EventBus`'s-or-a-singleton's static constructs first depends on runtime call order, not anything a caller controls. `EventBus`'s static could already be destroyed by the time a singleton's `ScopedSubscription`s destructed, and calling `EventBus::getInstance()` again on a dead function-local static is undefined behaviour.
- **Resolution:** Added a `bool g_eventBusAlive = true` (constant-initialized, safe to read from any other destructor regardless of order), flipped `false` in a newly-added `~EventBus()`, and checked by `ScopedSubscription::reset()` before it touches `EventBus::getInstance()`. Re-verified: `./verify_save_load` exits 0 across three repeated runs, ctest back to 13/13.

Issue 5: 28 `dynamic_cast`s in `CollisionResolver.cpp`

- **Problem:** Type dispatch by `dynamic_cast` chain outside the resolver's own ordered `EntityCategory`-pair switch had crept back to 28 hits in `CollisionResolver.cpp`, against a plan target of fewer than 4 — a live erosion of the R5 collision-dispatch refactor's Open/Closed intent (SPEC's own audit had separately flagged this as D8/A-10).
- **Resolution:** Replaced the cast chains with virtual contact hooks on `Entity` (`onStomped`, `onHitFromBelow`, `onPlayerTouch`, etc.) dispatched through the existing ordered `EntityCategory`-pair switch, per member_a_fix_plan.md WP12's acceptance bar. One dispatch family fixed per commit, with a full rebuild and full ctest run after each, per the orchestrating session's explicit discipline; a cast was kept, and documented as the

honest answer, wherever it was genuinely the simplest correct option rather than forced to zero for the sake of the count.

Issue 6: The ctest Suite Passed Vacuously Under a Release Build

- **Problem:** `CMAKE_BUILD_TYPE=Release` 's stock `-O3 -DNDEBUG` compiles every `assert()` in the `verify_*` harnesses (`verify_all` 54 assertions, `verify_enemies` 48, `verify_save_load` 32, `verify_blocks` 20, etc.) to nothing, so those targets printed `PASSED` and exited 0 without evaluating a single assertion — proven empirically during R5 by mutating `HiddenBlock` and watching the suite stay green.
- **Resolution:** Stripped `NDEBUG` from test targets only, at the `add_verify_test()` chokepoint in `SuperMarioGame/CMakeLists.txt` (`target_compile_options(${name} PRIVATE -UNDEBUG)`), deliberately choosing this over rewriting ~230 `assert()` call sites to a `check()` helper — the game binary itself keeps `NDEBUG` (assertions in shipped code stay compiled out, as intended). Verified by mutation: baseline the bug, fix it, re-break it, revert, confirming the guard actually fails when it should.

Issue 7: A Boss Pushed Out of Its Arena Falls Forever

- **Problem:** A user-reported live session (with screenshot) showed the Bowser health bar still on-screen while Bowser himself had fallen out of the level entirely, with no way to end the encounter — the boss had been physically knocked or walked past its arena's bounds with nothing to stop it falling indefinitely off-screen.
- **Resolution:** Added `Boss::returnToArenaSpawn()` in `Entities/Boss.cpp`, teleporting a boss that leaves its arena's bounds back to its spawn point instead of leaving it to fall forever. This same method became the template for the R21 batch's `DebugCheats` immortality rescue (Task 27 above), which explicitly cites it as the existing solution to "falls out of the playable area" that the player itself had never been given.

Issue 8: Map Editor Palette Silently Placed Nothing for 24 of 40 Buttons

- **Problem:** `PlaceEntityCommand` hand-rolled 16 type names in a `switch` and never called `EntityFactory::create` — the one path every other spawner in the codebase uses. The test that looked like it covered this actually called `EntityFactory::create` directly (the path placement did **not** use), while its own comment described this exact failure mode. Net effect: 24 of the editor's 40 palette buttons placed nothing, silently.
- **Resolution:** Rebuilt the editor as a real `Core::EditorState : IGameState` that places entities through `EntityFactory::create` (backed by the new `EntityCatalogue` registry from Task 25), closing the type-list drift permanently since the registry is now the single source both the factory and the editor read. Also closed a use-after-free found

during this work: `forgetEntity` handled the boss/shadow/player-2 raw pointers but not Player 1, so erasing Mario left three dangling raw pointers into `m_entities`.

Issue 9: Bowser's Fireball Stagger Was the Inverse of the Reported Bug

- **Problem:** Reported as "Bowser does not receive fireballs but can still be stomped." The enclosure-post hypothesis was tested and refuted both geometrically and empirically (a fireball registers hit on frame 0). The real defect was the **inverse** of the report: a bursting fireball stayed collidable through its full 0.24s explosion animation, so a single shot scored three hits against Bowser's stagger counter — invisible on an ordinary enemy, which dies to the very first hit. Because Bowser closes to melee range in roughly one second and every player-boss contact costs a form, and `onTookHit()` also zeroed the player's own fire-flower count on a successful stomp, the fight was winnable only by an undesigned route (repeated stomping) and effectively unwinnable by its designed one (four fireballs to stagger, then stomp).
- **Resolution:** Gated Bowser's stomp-vulnerability behind a stagger state exposed as a virtual on `Boss` rather than a `dynamic_cast` in the resolver, and fixed the fireball hit-window so a single burst can only register once. Paired with Lakitu now dropping a Fire Flower (Task 29) so the fight has a guaranteed, repeatable route to the fireballs the stagger requires even after a player has lost their power-up 114 tiles back. Verified live: fireballs 1→2→3→4 staggering Bowser with the HUD counting down to "STUNNED - STOMP NOW", and contact refused while he is guarded.

Issue 10: Two-Player Camera Froze After an Elimination

- **Problem:** The camera dispatch chose `updateVersusCamera` on `m_player2` alone whenever both players existed; when P2 was eliminated, that call returned early because it required both players, and the lone-player tether path had a mirrored version of the same bug — so the surviving player got neither camera. `updateBossArena` / `releaseBossArena` also read `m_player` directly rather than the active participant, so arena lock, escape clamp and battle music went dead for a 2P boss-fight survivor. Separately, P1's HUD badge went stale showing `setupTestScene`'s mock values in a real run, and P2's badge vanished entirely on elimination rather than showing an eliminated state.
- **Resolution:** Added `PlayingState::activeParticipant()` so camera, arena and music logic all resolve the live player rather than assuming both exist; added `rememberEliminatedIdentity()` / `applyEliminatedBadges()` to retain an eliminated player's HUD identity before `destroy()` nulls the pointer, so the badge can dim and stamp "OUT" instead of disappearing. Verified live: MARIO's badge dimmed with a red "OUT" while P2 continued playing and the terrain scrolled correctly behind the survivor.

Issue 11: A Flagpole Touched During a Boss Fight Permanently Disabled Itself

- **Problem:** `Flagpole::onPlayerCollision` latched `m_triggered`, paid the score, and published `LevelComplete` unconditionally on first touch. `PlayingState`'s handler then refuses that completion while a boss is alive (SPEC 6.4, "no escape until defeated") — but the latch stayed set regardless, so every later touch hit `if (m_triggered) return;` and published nothing. A player who brushed the flag mid-fight (level_2's Boom Boom arena spans tiles 176–192; the flagpole stands at 193, one tile past the fight's movement clamp) permanently spent the flagpole: boss dead, flag present, level impossible to finish. The refused touch also paid 5000 points and played the flagpole jingle, so it looked like something had succeeded, which is why one player reproduced the bug and the reporting teammate could not.
- **Resolution:** Both the collision handler and the gate now call one shared `levelMayComplete()` rather than maintaining two independent copies of "may the level end now" (a class of duplication this batch had already paid for more than once). A refused touch now leaves no trace at all — no latch set, no score paid, no jingle played — and `PlayingState` installs the completion gate on every `Flagpole` through `admitEntity`, the same single door the difficulty scale uses, so an editor-placed flagpole and a level-loaded one are gated identically. ctest 28/28 → 29/29 (new case in `tests/verify_r21_flagpole_softlock.cpp`).